

TDAPM: Target Distribution Alignment via Progressive Matching for Multi-Source-Free Domain Adaptation – *Supplementary Materials*

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In what follows, we provide a more detailed theoretical derivation for the progressive prediction alignment strategy employed in TDAPM.

SI. Theoretical Derivations for Progressive Prediction Alignment

We provide a detailed derivation of the prediction-shift stability bound presented in Section III.B of the main manuscript. Consistent with the main manuscript, we focus on the progressive adaptation trajectory from $\theta_{\mathcal{T}}^{(1)}$ to $\theta_{\mathcal{T}}^{(K)}$ and exclude the initialization transition from $\theta_{\mathcal{T}}^{(0)}$ to $\theta_{\mathcal{T}}^{(1)}$.

For each target input x , let $p^{(k)}(x) = \text{softmax}(\theta_{\mathcal{T}}^{(k)}(x))$ denote the predictive probability vector of $\theta_{\mathcal{T}}^{(k)}$. The prediction shift between two consecutive adaptation stages is defined as

$$\pi_k = \mathbb{E}_{x \sim P_{\mathcal{T}}^X} \left[D_{\text{SKL}} \left(p^{(k)}(x), p^{(k-1)}(x) \right) \right], \quad k = 2, \dots, K, \quad (1)$$

where $P_{\mathcal{T}}^X$ denotes the target-domain input distribution, and $D_{\text{SKL}}(p, q) = \frac{1}{2} [D_{\text{KL}}(p||q) + D_{\text{KL}}(q||p)]$ denotes the symmetric KL divergence. As defined in Eq. (13) of the main manuscript, the pairwise alignment loss at the k -th adaptation iteration is

$$\mathcal{L}_{\text{align}}^{(k)} = D_{\text{KL}} \left(p^{(k-1)}(x'_i) || p^{(k)}(x_i) \right), \quad (2)$$

where x'_i is the reliable matched sample of x_i selected using the preceding model $\theta_{\mathcal{T}}^{(k-1)}$. This loss encourages semantic consistency between the current prediction $p^{(k)}(x_i)$ and the reliable historical prediction $p^{(k-1)}(x'_i)$. Because Eq. (2) compares the predictions of two matched samples and employs the asymmetric KL divergence, it does not directly minimize the same-sample symmetric prediction shift π_k in Eq. (1). Instead, it provides an implicit mechanism for suppressing abrupt semantic changes during progressive adaptation.

Let $L(\theta_{\mathcal{T}}^{(k)}, P_{\mathcal{T}}) = \mathbb{E}_{(x,y) \sim P_{\mathcal{T}}} [\ell(p^{(k)}(x), y)]$ denote the target risk of $\theta_{\mathcal{T}}^{(k)}$. We assume that the clipped cross-entropy loss ℓ is L_{ℓ} -Lipschitz continuous with respect to the predictive probability vector under the ℓ_1 norm. Therefore, for any predictive probability vectors p and q ,

$$|\ell(p, y) - \ell(q, y)| \leq L_{\ell} \|p - q\|_1. \quad (3)$$

For two consecutive target models, we have

$$\begin{aligned} & |L(\theta_{\mathcal{T}}^{(k)}, P_{\mathcal{T}}) - L(\theta_{\mathcal{T}}^{(k-1)}, P_{\mathcal{T}})| \\ &= \left| \mathbb{E}_{(x,y) \sim P_{\mathcal{T}}} [\ell(p^{(k)}(x), y) - \ell(p^{(k-1)}(x), y)] \right| \\ &\leq \mathbb{E}_{(x,y) \sim P_{\mathcal{T}}} \left[|\ell(p^{(k)}(x), y) - \ell(p^{(k-1)}(x), y)| \right] \\ &\leq L_{\ell} \mathbb{E}_{x \sim P_{\mathcal{T}}^X} [\|p^{(k)}(x) - p^{(k-1)}(x)\|_1]. \end{aligned} \quad (4)$$

According to Pinsker's inequality, for any two probability vectors p and q , we have $\|p - q\|_1 \leq \sqrt{2D_{\text{KL}}(p||q)}$. By exchanging p and q , we also have $\|p - q\|_1 \leq \sqrt{2D_{\text{KL}}(q||p)}$.

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Therefore,

$$\begin{aligned}\|p - q\|_1 &\leq \sqrt{2 \min \{D_{\text{KL}}(p\|q), D_{\text{KL}}(q\|p)\}} \\ &\leq \sqrt{2D_{\text{SKL}}(p, q)}.\end{aligned}\tag{5}$$

Combining Eqs. (4) and (5) gives

$$\begin{aligned}&|L(\theta_{\mathcal{T}}^{(k)}, P_{\mathcal{T}}) - L(\theta_{\mathcal{T}}^{(k-1)}, P_{\mathcal{T}})| \\ &\leq \sqrt{2}L_{\ell}\mathbb{E}_{x \sim P_{\mathcal{T}}^x} \left[\sqrt{D_{\text{SKL}}(p^{(k)}(x), p^{(k-1)}(x))} \right].\end{aligned}\tag{6}$$

Since the square-root function is concave, Jensen's inequality yields

$$\begin{aligned}&\mathbb{E}_{x \sim P_{\mathcal{T}}^x} \left[\sqrt{D_{\text{SKL}}(p^{(k)}(x), p^{(k-1)}(x))} \right] \\ &\leq \sqrt{\mathbb{E}_{x \sim P_{\mathcal{T}}^x} [D_{\text{SKL}}(p^{(k)}(x), p^{(k-1)}(x))]} \\ &= \sqrt{\pi_k}.\end{aligned}\tag{7}$$

Consequently, the target-risk variation between two consecutive adaptation iterations satisfies

$$|L(\theta_{\mathcal{T}}^{(k)}, P_{\mathcal{T}}) - L(\theta_{\mathcal{T}}^{(k-1)}, P_{\mathcal{T}})| \leq \sqrt{2}L_{\ell}\sqrt{\pi_k}, \quad k = 2, \dots, K.\tag{8}$$

Using the telescoping decomposition and the triangle inequality, we obtain

$$\begin{aligned}&|L(\theta_{\mathcal{T}}^{(K)}, P_{\mathcal{T}}) - L(\theta_{\mathcal{T}}^{(1)}, P_{\mathcal{T}})| \\ &= \left| \sum_{k=2}^K [L(\theta_{\mathcal{T}}^{(k)}, P_{\mathcal{T}}) - L(\theta_{\mathcal{T}}^{(k-1)}, P_{\mathcal{T}})] \right| \\ &\leq \sum_{k=2}^K |L(\theta_{\mathcal{T}}^{(k)}, P_{\mathcal{T}}) - L(\theta_{\mathcal{T}}^{(k-1)}, P_{\mathcal{T}})| \\ &\leq \sqrt{2}L_{\ell} \sum_{k=2}^K \sqrt{\pi_k}.\end{aligned}\tag{9}$$

Define the cumulative prediction path from $\theta_{\mathcal{T}}^{(1)}$ to $\theta_{\mathcal{T}}^{(K)}$ as $S_K = \sum_{k=2}^K \sqrt{\pi_k}$. The corresponding target-risk variation therefore satisfies

$$|L(\theta_{\mathcal{T}}^{(K)}, P_{\mathcal{T}}) - L(\theta_{\mathcal{T}}^{(1)}, P_{\mathcal{T}})| \leq \sqrt{2}L_{\ell}S_K.\tag{10}$$

This result shows that the target-risk variation is controlled by the cumulative prediction path. A smaller S_K corresponds to a tighter upper bound on the target-risk variation. However, this stability bound does not guarantee monotonic risk reduction or improved target accuracy.

SII. Influence of the Number of Adaptation Iterations

We evaluate $K \in \{5, 10, 15, 20, 30, 50\}$ on the Office-Home dataset, where K denotes the total number of target-model adaptation iterations. Here, $\widehat{\pi}_k$ denotes the empirical estimate of the population-level prediction shift π_k defined in Eq. (1), and $\widehat{S}_K$ denotes the empirical counterpart of the cumulative prediction path S_K defined in Section III.B. The mean prediction shift $\bar{\pi} = (K-1)^{-1} \sum_{k=2}^K \widehat{\pi}_k$ measures the average prediction variation between consecutive iterations, while $\widehat{S}_K = \sum_{k=2}^K \sqrt{\widehat{\pi}_k}$ measures the accumulated prediction variation. The results are macro-averaged over four target domains and reported as mean $\pm$ standard deviation over three random seeds. As shown in Table SI, when K increases from 5 to 50, $\bar{\pi}$ decreases from 0.0218 to 0.0082, whereas $\widehat{S}_K$ increases from 0.5685 to 3.5528. Target accuracy increases from 75.1% at $K = 5$ to its highest observed value of 77.4% at $K = 15$. These results show that the mean one-step shift, cumulative prediction variation, and target accuracy exhibit different trends as K increases. Therefore, the iterative behavior cannot be characterized by K or the mean one-step prediction shift alone.

Table SI: Influence of the number of adaptation iterations on Office-Home. Results are reported as mean $\pm$ standard deviation over three random seeds and are macro-averaged over the four target domains.

K	Acc. (%)	Mean shift $\bar{\pi}$	$\widehat{S}_K$
5	75.1 $\pm$ 0.1	0.0218 $\pm$ 0.0006	0.5685 $\pm$ 0.0127
10	76.3 $\pm$ 0.1	0.0191 $\pm$ 0.0006	1.1605 $\pm$ 0.0210
15	77.4 $\pm$ 0.2	0.0167 $\pm$ 0.0003	1.6434 $\pm$ 0.0114
20	75.9 $\pm$ 0.4	0.0142 $\pm$ 0.0002	2.0152 $\pm$ 0.0123
30	75.8 $\pm$ 0.3	0.0115 $\pm$ 0.0002	2.6505 $\pm$ 0.0201
50	76.5 $\pm$ 0.3	0.0082 $\pm$ 0.0001	3.5528 $\pm$ 0.0468

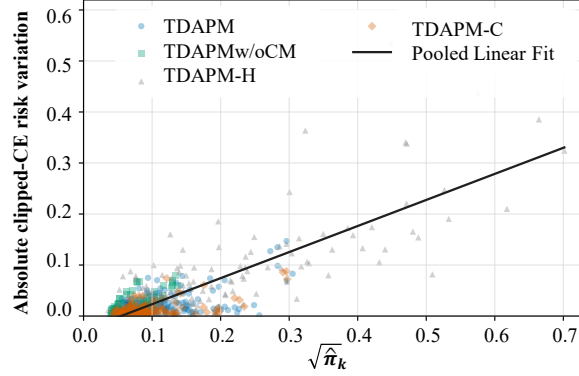


Fig. S1: Empirical relationship between $\sqrt{\widehat{\pi}_k}$ and $|\widehat{L}_k - \widehat{L}_{k-1}|$ on the Office-Home dataset. The solid line denotes the linear fit over all observations.

III. Empirical Analysis of the Prediction-Shift–Risk Relationship

We examine the empirical relationship between prediction shift and target-risk variation, where $\widehat{L}_k$ denotes the average clipped cross-entropy target risk at iteration k . As shown in Fig. S1, $\sqrt{\widehat{\pi}_k}$ is positively associated with the absolute target-risk variation $|\widehat{L}_k - \widehat{L}_{k-1}|$ collected from four settings (*i.e.*, TDAPM, TDAPMw/oCM, TDAPM-H, and TDAPM-C), four target domains, and iterations $k = 2, \dots, 15$. The Pearson and Spearman correlation coefficients are 0.8004 and 0.4973, respectively, indicating a strong positive linear association and a moderate positive monotonic association. These correlations are computed across iteration-level observations and provide empirical evidence consistent with the stability relationship derived in Section III.B. Prediction shift is therefore treated as a continuous measure of prediction variation, with a larger value corresponding to a looser stability bound rather than a violation of boundedness.